

Project Proposal

Introduction to Algorithms and Software Programming (SL 203)

Optimizing Cardiac Ward Workflows: An Algorithmic Approach to Dynamic Task Management and Allocation

Subash C, MBBS
subashc@iisc.ac.in

Course Instructor: Nagarajan Natarajan, PhD (Visiting Professor)
Course Instructor: Tarun Rambha, PhD (Associate Professor)

Indian Institute of Science, Bangalore

1 Objective:

This project aims to design and build an algorithm-based ward task management system that optimally schedules tasks to healthcare workers based on their skills, availability, and priority. This project focuses specifically on the cardiology ward, as a scoped implementation for a course project. The expected outcome is a demonstrably more optimised task allocation system that can be used in hospitals to improve the quality of care, minimize task turnaround time, and reduce the workload of healthcare workers.

2 Rationale:

Healthcare professionals have faced burnout for decades. Many hospitals in India are still using pen and paper to allocate tasks to healthcare workers. Doing so leads results in inefficient resource utilization. For example, the first task on the paper gets done first, irrespective of its priority.

3 Approach:

4 Timeline:

- **Week 1:**
 - To obtain a list of cardiovascular diseases and tests (to fetch it from the official website of the Mayo Clinic).
 - Categorize the diseases and tests based on whether the particular disease requires admission to the ward or not.
 - Identify the optimal data structure to store patient demographics, healthcare workers (Doctors, Nurses) data, diseases and tests, and the relationships between them.
- **Week 2:** [Milestone description]
- **Week 3:** [Milestone description]

5 Possible Issues:

[Write your possible issues here.]